Symmetries in Physics — Exercise Sheet 3

Representations and equivalence

Variant: Questions only

Exercise 1.

(i) Find all one-dimensional representations of D_n for $n \geq 3$, that is, all possible values of the numbers $x = \rho(r)$ and $y = \rho(s)$. How many are there when n is odd, and how many when n is even? *Hint*: numbers commute, so the relation $srs^{-1} = r^{-1}$ becomes an equation for x.

(ii) Give the kernel of each.

(iii) Show that none of them is faithful.

Exercise 2. Let D_4 act on the plane as on Sheet 1: r is the rotation through $\pi/2$ and $s(x_1, x_2) = (x_1, -x_2)$ the reflection in the x_1-axis, so

$$D(r) = R(\pi/2) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The square has its vertices at $q_1 = (1, 1)$, $q_2 = (-1, 1)$, $q_3 = (-1, -1)$ and $q_4 = (1, -1)$, and functions on the plane transform by the function-space representation

$$\big(\rho(g)f\big)(\vec{r}\,) = f(g^{-1}\vec{r}\,).$$

(i) Write down $r^{-1}(x_1, x_2)$ and $s^{-1}(x_1, x_2)$. Express the four functions $\rho(r)x_1$, $\rho(r)x_2$, $\rho(s)x_1$, $\rho(s)x_2$ as linear combinations of the coordinate functions x_1 and x_2.

(ii) Write the matrices of the operators $\rho(r)$ and $\rho(s)$ in the basis (x_1, x_2); column j lists the coordinates of the image of the j-th basis vector. Compare them with $D(r)$ and $D(s)$.

(iii) Evaluate the function $\rho(r)x_1$ at the vertex q_1. Find the vertex $r^{-1}q_1$, and check that $(\rho(r)x_1)(q_1) = x_1(r^{-1}q_1)$. What would the rule without the inverse, $f(g\vec{r}\,)$, have given?

Exercise 3. Let D_4 act on functions on the plane as in Exercise 2, now on the three-dimensional space of quadratic polynomials spanned by x_1^2, x_1x_2, x_2^2.

(i) Show that $\rho(g)(fh) = (\rho(g)f)(\rho(g)h)$ for any two functions f and h. Use this and Exercise 2 to write the matrices of the operators $\rho(r)$ and $\rho(s)$ in the basis (x_1^2, x_1x_2, x_2^2).

(ii) Find a basis in which $\rho(r)$ and $\rho(s)$ are both diagonal. The representation is then a direct sum of three one-dimensional representations: say which ones of Exercise 1 they are.

(iii) Which one-dimensional representation of D_4 does not occur?

Exercise 4. Let $C_2 = \{e, p\}$ act on wave functions $\psi \in L^2(\mathbb{R})$ by the parity operator $(\Pi\psi)(x) = \psi(-x)$, where $\Pi = \rho(p)$.

(i) Show that this is the function-space representation of the action $p \cdot x = -x$ of C_2 on $\mathbb{R}$.

(ii) Show that $\Pi^2 = \mathbb{1}$, and deduce that the only possible eigenvalues of Π are $+1$ and -1. Show that the eigenfunctions are exactly the even functions (eigenvalue $+1$) and the odd functions (eigenvalue -1).

(iii) Prove that every ψ is, in exactly one way, the sum of an even and an odd function, so that $L^2(\mathbb{R})$ is the direct sum of the even and the odd functions. How does p act on each summand?

Exercise 5. Let $C_3 = \langle r \rangle$ act on itself by left multiplication, and number the basis of its regular representation as $e_1 = e_e$, $e_2 = e_r$, $e_3 = e_{r^2}$.

(i) Write the matrices of the operators $\rho(e)$, $\rho(r)$, $\rho(r^2)$ in this basis.

(ii) Show that the representation is equivalent to a direct sum of one-dimensional representations. Exhibit the change of basis, and say which of the χ_k of Lecture 3 occur. *Hint*: an eigenvector of $D(r)$ is automatically an eigenvector of $D(r^2) = D(r)^2$.

Exercise 6. Lecture 3 writes the defining representation of D_3 in a basis in which s is the reflection in the y-axis. On Sheet 1, s was the reflection in the x-axis. With $R(\theta)$ the rotation through θ, the two versions are

$$D(r) = R(2\pi/3), \quad D(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \qquad \text{and} \qquad D'(r) = R(2\pi/3), \quad D'(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

(i) Show that the quarter turn $T = R(\pi/2)$ satisfies $T\,D'(g) = D(g)\,T$ for $g = r$ and $g = s$, and explain why checking the two generators is enough. So the two versions are equivalent.

(ii) Use Exercise 1 to show that the defining representation is not equivalent to any direct sum of two one-dimensional representations of D_3.

(iii) The rotations alone can be split: Lecture 3's last example makes every $D(r^k)$ diagonal in the basis $v_\pm = \binom{1}{\mp \mathrm{i}}$. Write $D(s)$ in that basis. What does the reflection do to the lines $\mathbb{C}v_+$ and $\mathbb{C}v_-$? Conclude that no line of $\mathbb{C}^2$ is mapped to itself by both $D(r)$ and $D(s)$.